WELLSENSE · RISK REGISTER

WellSense risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade) with the project risk register, preliminary hazard analysis and first-pass design FMEA of the current design (option A)

WellSense risk register and hazard analysis

The top project risks are evidence for drinking-water safety (no material certificates for the low-cost wetted parts) and whether a low-cost vented probe holds ±20 mm and 20 mm a year in a real well; the top hazards are contaminating a supply well and electric shock from the existing pump's mains supply while the wellhead is open. The design FMEA's highest RPN items are vent blockage (F-1) and transducer drift (F-4), both 36, then a gap in the seal plate (F-9, 30); the tube string dropping into the well (F-2) scored 60 before the clevis pin of WLS-DEC-001 D-S1 and 20 after it. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found) to 5 (unlikely to be found).

> **Safety:** Drinking water, mains-powered pumps, open wells, a lithium iron phosphate cell in the FieldNode core and heavy cable runs. See WLS-PRC-001, Safety, and the safety stops in WLS-BLD-001.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R-1	No drinking-water certificates for wetted parts	Low-cost suppliers rarely hold NSF/ANSI/CAN 61 or equal for probes, cable and seal materials	PR-009 stays not met; partner or authority will not allow the probe in a supply well	4	5	Design lead	Ask suppliers at purchase (DEC to confirm item 1); fall back to certified grades and accept the cost; option C reduces wetted parts to the tube (WLS-OPT-001)
R-2	Low-cost probe misses ±20 mm or drifts more than 20 mm a year	Unmeasured nonlinearity, thermal effect and drift of low-cost transducers	PR-004 and PR-005 not met; data not trusted by the community	3	4	Design lead	Side-by-side tape record at TRL 4; quarterly tape checks correct drift; choose the 0.25 % class
R-3	Access tube cannot pass an installed pump's riser and cable	Tight casing, couplings and cable clamps on the riser	Tube only fits when the pump is pulled, which owners avoid	3	4	Partner field lead	Installation trial with the partner; fit the tube when a pump is next pulled; a 19 mm probe gives more room
R-4	Cost stays over the USD 200 target	Locks, conduit, constructable wellhead parts	PR-012 over target (USD 64.40 now)	4	2	Project lead	Value-engineering savings in WLS-DEC-001; option C if proven; budget is a control target, not a limit
R-5	Partner not engaged	ACWADAM not yet approached	Requirements stay the author's reading; no field trial	3	4	Project lead	Approach the first candidate partner (WLS-DEC-001); Helpful Engineering network as fallback
R-6	FieldNode core not confirmed at 55 °C ambient	FieldNode rated to 45 °C with its shield	Hot sites excluded (PR-015 at risk)	3	3	FieldNode project	Sites above 45 °C not used until FieldNode confirms (WLS-DEC-001)
R-7	Dashboard not built	Software not started	PR-013 not met; data not used in meetings	3	3	Software lead	Plan it in co-design (WLS-PLN-001); CSV export first
R-8	Data misuse or conflict over shared groundwater	Level data could be used against a community or a farmer	Loss of trust, removal of loggers	2	4	Partner	Levels only; community controls sharing (PR-014); agree rules before installing

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H-1	Contamination of a drinking water supply	Undisinfected probe, cable or tube; wetted materials that leach; wellhead left open	Illness of everyone drinking from the well	5	2	Disinfect everything lowered in (AWWA C654 practice, STM S25); certified materials (PR-009); split seal plate with gasket and wraps; install with the owner or authority	Medium until certificates are in hand
H-2	Electric shock from the pump's mains supply	Working at an open wellhead beside a live pump cable	Electric shock, death	5	2	Isolate and lock off the pump supply before opening the wellhead (build plan safety stops); WellSense itself is 24 V or less (PR-018); logger wiring kept apart from the pump cable	Low
H-3	Fall or drowning at an open well	Large-diameter or open well left uncovered during work	Injury or death, especially of children	5	1	Never leave a wellhead open; cover large wells while working (PRC Safety)	Low
H-4	Tube string, probe or cable dropping into the well	Collar grip creeps on PVC; cable runs away while lowering	Lost probe, pump fouled, well out of use	3	2	Clevis pin on the collar (D-S1, STM S14); two people lower the string slowly (STM S26); cable strain member factor 11.5	Low
H-5	Lithium iron phosphate cell fault in the FieldNode core	Charging out of temperature, damage	Fire, burns	4	1	FieldNode's fusing and charge limits; hot-site rule (WLS-DEC-001)	Low
H-6	Manual handling strain	Lifting the post, footing concrete, a 60 m tube string	Back injury	2	2	Two people; loads under 25 kg each (STM S26)	Low
H-7	Cuts and swarf	Drilling PVC, HDPE and aluminium; sawing the post	Cuts, eye injury	2	3	Eye protection, deburring (WLS-BLD-001 section 7)	Low
H-8	Tampering or theft at the wellhead	Exposed parts at a public water point	Contamination path, lost logger	3	3	Locks on the seal plate, cap and box; cable in conduit	Low to medium

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
						(PR-016); steel cover at exposed sites	

Design FMEA (first pass, P3)

Table 3. Failure modes of the chosen design (option A). RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F-1	Vent capillary and breather (BOM 2, 6)	Keep the transducer referenced to air	Water or condensate blocks the vent	Reading drifts with air pressure by up to ±306 mm	Desiccant spent; condensation in the cable in a cool well	4	3	3	36	Quarterly desiccant change (PR-017, 552 days of capacity); barometric sensor flags a blocked vent by correlation
F-2	Collar, clevis pin and seal plate (BOM 4)	Carry the access tube string	Tube slips through the collar	String drops into the well, probe lost	PVC creep under the collar clamp	5	1	4	20	Closed by D-S1: clevis pin takes the load if the grip slips (was O 3, RPN 60 before the fix)
F-3	Access tube joints (BOM 3)	Let the probe run freely	Probe snags on a cement bead	Probe cannot be lowered or lifted for calibration	Excess solvent cement; thick-walled pipe	3	2	2	12	Closed by D-S2: wipe joints, pull a 23 mm drift through before the probe
F-4	Transducer (BOM 1)	Read the head of water	Offset or span drift	Level error beyond ±20 mm	Low-cost sensor ageing, temperature	4	3	3	36	Quarterly tape check and correction (PR-005); 0.25 % class probe with a factory calibration certificate for span (D-S3); side-by-side record at TRL 4
F-5	Interface board, boost (BOM 7)	Power the loop at 24 V	Boost fails or rail sags	No reading	Component failure, surge	3	2	1	6	FieldNode reports a missing reading at once; TVS surge protection; spare module costs a few dollars
F-6	Shunt and ADC (BOM 7)	Convert current to a number	Gain drift with box temperature	Level error up to a few millimetres	10 ppm/K shunt, ADC drift over 30 K	2	3	4	24	0.1 % 10 ppm/K shunt; drift in the error budget (WLS-CAL-001 [C3])

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F-7	Vented cable (BOM 2)	Carry signal and vent; hold the probe	Jacket cut or crushed	Water ingress, lost reading, vent blocked	Livestock, mowing, tampering at the surface	4	2	2	16	Galvanized conduit and flexible tail (PR-016)
F-8	Cable support grip and eye bolt (BOM 5)	Set and hold the probe depth	Grip slips	Probe depth changes, offset error	Grip not suited to the cable diameter	3	2	3	18	Grip for 6 to 8 mm cable, 35 N (DEC to confirm item 5); the tape check finds an offset
F-9	Seal plate, gasket and wraps (BOM 4)	Keep surface water and insects out	Gap at the riser or pump cable	Contamination path into the well	Riser or cable diameter differs from design	5	2	3	30	Measure the riser and cable at the well before cutting the plate (DEC to confirm item 2); EPDM wraps take up a few millimetres
F-10	Junction box and glands (BOM 6)	Keep the electronics dry	Water ingress	Corrosion, lost reading	Gland not tightened; lid seal aged	3	2	3	18	IP66 box; glands checked each quarter
F-11	FieldNode lead and M12 plug (BOM 15)	Carry power and I2C	Plug loosened or corroded	No reading	Vibration, moisture	3	2	1	6	IP67 plug; missing readings show at once
F-13	Interface board, shunt (BOM 7)	Convert loop current	Shunt burns out	No reading; board replaced	Short between the cable cores puts 24 V across the shunt (149 mA, 3.3 W)	3	2	2	12	Closed by D-S4: 50 mA resettable fuse in the loop (was O 2 with no protection, RPN 12 with a board lost)
F-12	Post and footing (BOM 10, 13)	Hold the logger	Post leans or is knocked over	Conduit strained, logger damaged	Livestock, soft ground	3	2	1	6	0.6 m concrete footing; visible at each visit

Option notes

Options B and C (WLS-OPT-001) would change F-1, F-4 and F-7: B replaces the vent with a pump and valve (new failure modes: pump diaphragm wear, line leak, water entering the line during an outage), and C removes every wetted electronic part (new failure modes: condensation on the head's speaker and microphone, insects in the tube, lost echo at depth). These are not scored until Amish chooses (WLS-DDR-005).